Librito Definiciones

Estadística

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Definiciones

n = tamaño de la muestra

Parametro	Estadistico
μ = media poblacional	$\bar{x}$ = Media muestral
σ^2 = Varianza Poblacional	s^2 = Varianza Muestral = Cuasivarianza muestral
σ = Desviación Tipica Poblacional	s = Desviación Tipica Muestral
π = Proporción Poblacional	p = proporción muestral

Intervalos de Confianza

μ: Normal o t de Student

Caso 1: σ / σ^2 conocida

$$\mu \in \left(\bar{x} \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right)$$

Caso 2: σ / σ^2 desconocida, $n > 40$

$$\mu \in \left(\bar{x} \pm z_{1-\alpha/2} \cdot \frac{s}{\sqrt{n}} \right)$$

Caso 3: σ / σ^2 desconocida, $n \leq 40$

$$\mu \in \left(\bar{x} \pm t_{n-1,\,1-\alpha/2} \cdot \frac{s}{\sqrt{n}} \right)$$

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σ^2 ; χ^2 → Chi Cuadrado

$$\sigma^2 \in \left(\frac{S^2(n-1)}{\chi^2_{n-1,\frac{\alpha}{2}}} ; \frac{S^2(n-1)}{\chi^2_{n-1,1-\alpha/2}} \right)$$

π : Normal

$$\pi \in \left(p \pm z_{1-\alpha/2} \cdot \sqrt{\frac{p(1-p)}{n}} \right)$$

$\mu_1 - \mu_2$: Diferencia de Medias

Caso 1: σ_1 y σ_2 conocidas

$$\mu_1 - \mu_2 \in \left(\bar{x}_1 - \bar{x}_2 \pm z_{1-\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right)$$

Caso 2: σ_1 y σ_2 desconocida, n_1 y $n_2 > 40$

$$\mu_1 - \mu_2 \in \left(\bar{x}_1 - \bar{x}_2 \pm z_{1-\alpha/2} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right)$$

Caso 3: σ_1 y σ_2 desconocida, n_1 o $n_2 \leq 40$

$$\mu_1 - \mu_2 \in \left(\bar{x}_1 - \bar{x}_2 \pm t_{\epsilon, 1-\alpha/2} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right) \rightarrow \epsilon = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

Si $\sigma_1 = \sigma_2 \rightarrow \epsilon = n_1 + n_2 - 2$

$\pi_1 - \pi_2$: Diferencia de Proporciones

$$\pi_1 - \pi_2 \in \left(p_1 - p_2 \pm z_{1-\frac{\alpha}{2}} \cdot \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} \right)$$

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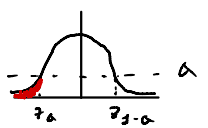
Test de Hipótesis

H_0: Hipótesis Nula

H_1: Hipótesis Alterna : H_A

Caso 1:

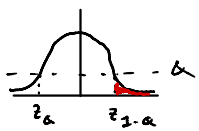
$$\begin{cases} H_0: \mu = x \\ H_1: \mu < x \end{cases}$$



■ -> Región Crítica

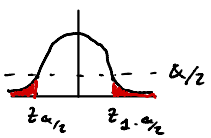
Caso 2:

$$\begin{cases} H_0: \mu = x \\ H_1: \mu > x \end{cases}$$



Caso 3:

$$\begin{cases} H_0: \mu = x \\ H_1: \mu \neq x \end{cases}$$



μ:

Caso 1: $z_{obs} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ Caso 2: $z_{obs} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ Caso 3: $t_{obs} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$

σ^2: $\chi^2_{obs} = \frac{s^2(n-1)}{\sigma^2}$

π: $z_{obs} = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$

$\mu_1 - \mu_2$:

Caso 1: $z_{obs} = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ Caso 2: $z_{obs} = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

Caso 3: $t_{obs} = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

$\pi_1 - \pi_2$: $z_{obs} = \frac{p_1 - p_2 - (\pi_1 - \pi_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} - \frac{p_2(1-p_2)}{n_2}}}$

Si $z_{obs} / t_{obs} / \chi^2_{obs} \in RC$
-> Rechazo H_0

Si: $z_{obs} / t_{obs} / \chi^2_{obs} \notin RC$
-> No puedo rechazar H_0

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